MIL-C-22764C(WP)
21 OCTOBER 1964
SUPERSEDING
MIL-C-22764B(WEP)
1 AUGUST 1963

MILITARY SPECIFICATION

CODER-RECEIVER-TRANSMITTER KY-532 (*)/ASQ

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements of the Coder-Receiver-Transmitter KY-532(*)/ASQ. When used as a part of Integrated Electronic Centrals AN/ASQ-19, AN/ASQ-82 and AN/ASQ-88 or other centrals, it shall provide the following:

(1) Reception of interrogating signals (R-F Pulses).

(2) Decoding of the interrogating signals.

(3) Generation of appropriate codes for the modes of interrogation.

(4) Transmission of the appropriate code to an antenna terminal.

1.2 Associated Equipment - The equipment shall be capable of operating with the following associated equipment which is not to be supplied as part of this equipment.

Item	Type Designation
Amplifier-Power Supply-Receiver	AM-2349/ASQ-19
Control, Radar Set	C-1159/APX-6B
Control, Radar Set	C-1272A/APA-89 or C-1128/APX-25
Antenna	AS-133/APX or AT-234/APX
Control, Transponder Set	C-6280(P)( )/APX
Computer, Transducer Altitude, Altitude Encoding	CPU-46/A or equivalent

FSC 5841

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Computer, Transponder	KIT-1/TSEC
Integrated Electronics Central	AN/ASQ-19 ( ) or AN/ASQ-82( ) or AN/ASQ-88( )

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-I-18309	Integrated Electronics Central AN/ASQ-19
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-C-27889	Computer, Transducer, Altitude, Altitude Encoding, General Specification for
MIL-C-38257	Control, Transponder Set C-6280 (P)( )/APX

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STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft Characteristics and Utilization of

Federal

FED-STD-595 Colors

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (see 6.5)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

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3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship.

3.2.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 29.5 pounds.

3.2.2 Reliability -

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 600 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.2.2.3 Reliability In Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 325 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.2.4 Time Totalizing Meter - The equipment shall contain time a totalizing meter in accordance with Specification MIL-M-7793.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

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3.2.4 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment supplies or designated by the procuring activity.

3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181D.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($25^{\circ}C \pm 5^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Voltage	115 ± 1.0 V AC and 27.5 ± 0.5 V DC

3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class II equipment when mounted in AN/ASQ-19, AN/ASQ-82 and AN/ASQ-88 except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.10 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 5 minutes at extreme service conditions.

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3.2.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 80 VA

(2) DC Power 28V, Category "B", 200 ma.

3.2.11.1 Switched DC and AC Output - In addition to the above, provisions shall be made for switched routing of + 28 V DC and 115 V AC power to external pin connections. The d-c is used to energize the Mode 4 caution light circuitry (para 3.3.12.3) and reply light (para 3.3.12.2). The a-c is necessary to supply electrical loads requiring a maximum of 1.0 amperes. The two d-c loads will not exceed 100 ma. each. The a-c circuit shall be protected by an external circuit breaker and shall be controlled in such manner that it is supplied when the equipment is placed in any operating condition. Power shall not be supplied when the equipment is in the "off" condition.

3.3 Performance

3.3.1 General - Unless otherwise specified, the values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme service conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions are specified herein.

3.3.1.1 Operation - This equipment shall receive coded interrogations, shall decode these interrogations and, in response, shall transmit pulse coded replies. The reply and suppression characteristics specified herein shall apply from MTL to 50 db above MTL (3.3.3.2).

3.3.2 Conditions of Performance - Unless otherwise stated, performance requirements defined herein are based on the following conditions.

3.3.2.1 Interrogation Characteristics - Interrogation characteristics shall consist of 2 pulses in accordance with the following table:

Interrogation Mode	Pulse Separation	Rise Time	Pulse Width	Decay Time
1	3 ± 0.2 us	0.05 to 0.1 usec	0.8 ± 0.1 usec	0.05 to 0.2 us
2	5 ± 0.2 us	0.05 to 0.1 usec	0.8 ± 0.1 usec	0.05 to 0.2 us
3/A	8 ± 0.2 us	0.05 to 0.1 usec	0.8 ± 0.1 usec	0.05 to 0.2 us
C	21 ± 0.2 us	0.05 to 0.1 usec	0.8 ± 0.1 usec	0.05 to 0.2 us
4	**See Note	0.05 to 0.1 usec	0.5 ± 0.1 usec	0.05 to 0.2 us

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**Note: Four pulses referenced from the first pulse in multiples of 2 microseconds. Each pulse should be located within ± 0.1 usec from the referenced position. The four pulses may be followed by as many as 33 additional pulses, spaced as close as 2.0 usec.

3.3.2.2 Interrogation Frequency - The interrogation frequency will be 1030 ± 0.2 mc.

3.3.3 Receiver Characteristics

3.3.3.1 Receiver Frequency - The receiver shall be adjusted by the supplier such that the center of the receiver bandpass (between points 6 db down) is 1030 ± 0.5 mc.

3.3.3.1.1 Receiver Frequency Drift - Under all specified service conditions the frequency drift shall not exceed ± 1.5 MC.

3.3.3.2 Minimum Triggering Level (MTL) - Minimum triggering level is a term applied to the minimum amplitude of the input RF interrogation pulses which just cause full firing of the transmitter. The minimum triggering level measured at the receiver antenna receptacle under standard test conditions at any frequency to which the receiver is tuned shall be at least -77 dbm when the interrogation signal is supplied by a 50 ohm signal generator. The minimum triggering level shall not decrease to less than -74 dbm under any combination of service conditions as specified in Specification MIL-E-5400 for Class II equipment.

3.3.3.3 Receiver Triggering Level Selection - (Sensitivity) Provisions shall be included for remote triggering level selection as follows:

(1) "Normal" triggering level adjustable from at least -69 dbm to -77 dbm.

(2) "Low" triggering level adjustable from at least 12 to 25 db below that level of adjustment in 3.3.3.3 (1).

3.3.3.4 Random Triggering Rate or Squitter - The equipment shall not transmit more than 5 pulse trains per second averaged over one minute under the following conditions:

(1) The receiver input is terminated in a 51 ohm resistive load.

(2) No external signal is present.

(3) The triggering level control is selected to "Normal."

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3.3.3.5 Receiver Bandwidth - Under standard conditions, with interrogation pulse widths of 0.4 to 0.9 usec. duration the receiver bandwidth shall be as follows:

(1) At 6 db down - 8 ± 1 mc

(2) At 60 db down - not more than 50 mc

3.3.3.5.1 Receiver Bandwidth Drift - The overall receiver bandwidth under all combinations of service conditions shall not vary more than ± 3 mc in overall bandwidth at 6 db down from maximum response. However, in no case shall the overall bandwidth be less than 7 mc at points 6 db down from the maximum response.

3.3.3.6 I-F Frequency - Under standard conditions the I-F mid-frequency measured between points 6 db down from a maximum response shall be 60 ± 1.5 mc.

3.3.3.7 Rejection of Undesired Signals

3.3.3.7.1 I-F Rejection - RF input signals at the I-F amplifier frequency having amplitudes up to 60 db above that of the desired signal shall in no way degrade the specified performance of the equipment.

3.3.3.7.2 Image Rejection - RF input signals at the receiver image frequency having an amplitude up to 60 db greater than that of the desired signal shall in no way degrade the specified performance of the equipment.

3.3.3.7.3 RF Rejection - With the receiver tuned to 1030 mcs per second, the response to c-w signals, as measured by the voltage at the detector output, shall be at least 70 db below the response at the desired frequency at all frequencies lower than 1010 and higher than 1050 megacycles per second with the exception of the image frequencies which shall be at least 60 db below the response at the desired frequency.

3.3.3.7.4 Anti-Jamming Characteristics - The equipment shall operate under all service conditions in the presence of CW, AMCS, random-spaced pulses and regularly-spaced pulses of any spacing including radar interference as follows:

(1) If the interrogating signal to which the equipment should reply is between 10 and 50 db above the minimum triggering level, and any interfering signal 10 db down from the interrogating signal shall not reduce the number of valid replies by more than 50%.

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(2) For test purposes, the AMCW jamming signal may be simulated using a signal generator which can be modulated 30% by a 1000 cps sine wave; random spaced pulses shall not exceed a total of 1000 pps; regularly spaced pulses shall not exceed 400 pps.

(3) Mode 4 Sync Jamming - Mode 4 operation will require the use of anti-interference interrogation (AII) type signals. Recognition of this type of interrogation for sync repetition interference reduction will be accomplished in the receiver-transmitter. This recognition shall be accomplished by sensing the lack of any pulse in the Mode 4 interrogation for intervals of a minimum of 4 usec(4.5 ± 0.5 usec). An interrogation containing such gaps with neither an AII nor information pulse shall cause the generation of a Mode 4 computer reset trigger. Also, this trigger shall cause termination of the Modes 1, 2, 3/A and C, suppression within 2 usec after its receipt.

(4) If the interfering signal consists of randomly spaced pulses only and the interrogating signal is a Mode 1, 2, 3/A or C interrogation that is 10 db above the interfering signal, the equipment shall reply to at least 90 percent of the interrogations where the randomly spaced pulses are not recognized as normal or side lobe interrogations.

3.3.3.7.5 Input Protection - The equipment shall not be damaged by the application of an R.F signal of any frequency of any amplitude up to 1/2 volt and of any type of modulation applied and measured at the antenna terminal of the Receiver-Transmitter.

3.3.3.7.6 Pulse Width Discrimination - Modes 1, 2, 3/A and C - Received signals between minimum triggering level and 6 db above this level whose duration is less than 0.3 usec, shall not cause the transponder to initiate reply or suppression action. With the exception of pulses with amplitude variations approximately a normal interrogation pulse pair condition, any pulse or pair of pulses each of a duration greater than 1.5 usec shall not cause the transponder to initiate reply or suppression action.

3.3.3.7.7 Pulse Width Discrimination Mode 4 - Received signals between minimum triggering level and 6 db above this level whose duration is less than 0.2 usec. shall not cause the transponder to initiate reply or suppression action.

3.3.3.8 Automatic Overload Control (AOC) - The AOC shall operate to reduce the triggering level to an extent which will prevent the transmitter duty cycle from being exceeded. The AOC action shall be equally effective in all modes of operation. When operating within a 0.01 duty factor, the triggering level shall not be reduced by more than 3 db above MTL. When operating at a duty factor exceeding 0.012, the triggering level shall be reduced at least 30 db above MTL.

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3.3.3.8.1 Reply Rate Limiting - This control shall limit the number of generated reply trains in Modes 1, 2, 3/A, and C without regard to the number of pulses in each reply, or whether replies are normal, I/P or Emergency. Trigger-level reduction shall be gradual to prevent weaker interrogations from denying transponder service to stronger interrogations as a result of over-interrogation. Maximum reply rate shall not exceed 120% of the rate at which reduction first begins if the interrogation signal is increased to 50 db above minimum triggering level. Maximum reply rate shall be adjustable from 500 to 3000 per second.

3.3.3.8.2 Mode 4 Rate Limiter - See paragraph 3.3.11.4

3.3.3.9 Suppression Characteristics

3.3.3.9.1 Recovery Time, Internal Suppression - The sensitivity of the equipment when operating on Modes 1, 2, 3/A, or C shall be reduced to at least 69 db below the "Normal" minimum triggering level for 30 microseconds after the first transmitted pulse and shall remain at least 30 db below the "Normal" minimum triggering level for 50 microseconds after the first transmitted pulse, returning to within 3 db of "Normal" minimum triggering level within a period of 75 to 125 microseconds after the last reply pulse transmitted. The sensitivity of the equipment when set to transmit Emergency replies shall be reduced at least 69 db below the "Normal" minimum triggering level for 100 microseconds and shall return to within 3 db of the normal triggering level within a period of 100 to 125 microseconds after the last reply pulse transmitted. Action of the internal circuit that produces this desensitized period shall produce no interference or adverse effect on the functioning of the external suppression circuit described in 3.3.3.9.3. Recovery time shall be measured from the leading edge of the pulse of the first reply to the leading edge of the first pulse of the second reply at the 50% amplitude points of the pulse concerned.

3.3.3.9.2 Suppression During Mode 4 Interrogation - Decoding of Modes 1, 2, 3/A, and C, shall be suppressed when the first 4 pulses of a Mode 4 interrogation are received. Suppression due to a Mode 4 interrogation shall continue for at least 75 but not more than 125 microseconds after the second pulse, unless this suppression period is terminated by a disparity pulse (3.3.12.4). The first 2 pulses of a Mode 4 interrogation shall also be recognized as a legitimate ISLS control signal, and processed as required under 3.3.5.

3.3.3.9.2.1 Suppression During Transmission of Mode 4 Replies - Decoding of Modes 1, 2, 3/A, and C shall be suppressed during transmission of Mode 4 replies. Suppression shall continue for 35 $\pm$ 10 microseconds after the last transmitted pulse.

3.3.3.9.3 Suppression From External Equipment - The equipment shall provide for suppression from external sources having suppression pulse characteristics specified below. One input connector (which may be part of a main chassis connector) shall be provided for this suppression.

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(1)	Amplitude	15 to 70 volts
(2)	Polarity	Positive
(3)	Duration (measured at 15 volts amplitude)	1.5 to 250 microseconds
(4)	Input resistance	2200 ohms shunted by 500 picofarads. Damage to the equipment shall not result from an "open" or "short" of the input jack.
(5)	Rise Time	Equal to or greater than 10 volts per microsecond.
(6)	Decay Time (from 15 to 3 volts)	Not greater than 10 microseconds.

3.3.3.9.3.1 Operation - Upon application to the receiver-transmitter of a suppression pulse having the above characteristics, decoding of Modes 1, 2, 3/A, C, and 4 shall be suppressed for an interval starting not more than 1 microsecond after the suppression pulse has reached an amplitude of 10 volts and shall continue for at least the duration of the suppression pulse. After the end of the suppression pulse, the transponder shall regain normal receiver sensitivity, within 3 db, no later than 15 microseconds after the suppression pulse has fallen to 1.0 volt.

3.3.4 Decoder - The equipment shall incorporate decoding circuits for the purpose of distinguishing each of the 5 modes of interrogation described in 3.3.2.1 and shall furnish a trigger output to the appropriate mode coder. This circuitry shall be such that the reply transmitted by the equipment shall always correspond to the mode of interrogation.

3.3.4.1 Decoder Tolerance - Mode 1 - The decoder shall not cause the equipment to generate a Mode 1 reply to a double-pulse interrogation when the pulses are spaced less than 2.5 microseconds or greater than 3.8 Microseconds.

3.3.4.2 Decoder Tolerance - Mode 2 - The decoder shall not cause the equipment to generate a Mode 2 reply to a double-pulse interrogation when the pulses are spaced less than 4.2 microseconds or greater than 5.8 microseconds.

3.3.4.3 Decoder Tolerance - Mode 3/A - The decoder shall not cause the equipment to generate a Mode 3 reply to a double-pulse interrogation when the pulses are spaced less than 7.2 microseconds or greater than 8.8 microseconds.

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3.3.4.4 Decoder Tolerance - Mode C - The decoder shall not cause the equipment to generate a Mode C reply to a double-pulse interrogation when the pulses are spaced less than 20.2 microseconds or greater than 21.8 microseconds.

3.3.4.5 Decoder Tolerance - Mode 4 - Four pulse interrogation signal with pulses spaced within ± 0.1 usec from nominal position shall cause the equipment to generate Mode 4 enable triggers. Pulses spaced greater than ± 0.6 usec from nominal positions shall not cause such triggers.

3.3.4.6 Echo Rejection - The radio receiver-transmitter shall contain echo suppression circuitry designed to permit normal operation in the presence of echoes of signals in space.

3.3.4.6.1 Desensitization - Upon receipt of any pulse more than 0.7 microseconds in duration or group of four pulses each more than 0.4 microseconds in duration and spaced 2 microseconds, the sensitivity of the receiver shall be lowered so that an additional pulse 9 db or more below the amplitude of the single pulse or the four pulse group shall not be detected by the Mode 1, 2, and 3/A or Mode 4 decoders respectively. Upon receipt of a group of four pulses each more than 0.4 microseconds in duration and spaced 4 microseconds, the sensitivity of the receiver shall be lowered so that an additional pulse 12 db or more below the amplitude of the 4 pulse group and spaced 2 microseconds after the last pulse of the four pulse group should not be detected by the Mode 4 decoder. With the exception of the first microsecond following the trailing edge of the last desensitizing pulse, the sensitivity of the receiver shall not be lowered so that an additional pulse at the same amplitude as the single pulse or the four pulse group is not detected by the Mode 1, 2, and 3/A or Mode 4 decoders respectively.

3.3.4.6.2 Recovery - Following desensitization, the receiver shall recover sensitivity (within 3 db of minimum triggering level) within 15 microseconds after reception of a desensitizing pulse having a singal strength up to 50 db above minimum triggering level.

3.3.4.7 Decoder - Single Pulse - The equipment shall not respond to a single-pulse interrogation signal of any duration, provided that

(1) The pulse carries no ripples comparable to normal interrogations

(2) No accompanying CW signal capable of producing a beat frequency compatable to normal interrogation pulse spacing is present, and

(3) Pulse width does not correspond to interrogation pulse spacing when the single pulse is off frequency.

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3.3.4.8 Decoder Control - The equipment shall be such that operation with Control C-6280(P)( )/APX will permit selection of any combination of the following conditions of operation:

(1) Mode 1
(2) Mode 2
(3) Mode 3/A
(4) Mode C
(5) Emergency Operation
(6) I/P Operation

Operation with Control Radar Set C-1159/APX-6B and C-1272A/APA-89 provides discrete control of the decoding of Modes 2 and 3/A whereas Mode 1 is always enabled and Mode C and 4 are always disabled.

3.3.5 Side Lobe Suppression

3.3.5.1 Condition for Performance - The following defined characteristics are necessary for equipment operation in a three-pulse interrogation side lobe suppression system. For conditions applicable to Mode 4 see paragraph 3.3.12.4.

3.3.5.1.1 Transmitted Interrogation Control Pulse - The transmitted interrogation control pulse has the following characteristics:

3.3.5.1.1.1 Power - The control pulse power level is radiated omni-directionally and the power level is equal to or greater than the level of the greatest side lobe of the associated interrogation from a directional antenna. The omni-directional level is at least 13 db below the peak level of the directional antenna main lobe pulse.

3.3.5.1.1.2 Frequency - The frequency of the omni-directional control pulse transmission used has the same frequency tolerance as the interrogation pulse pair, but its frequency does not differ from that of the directional interrogation pulses by more than $\pm$ 0.2 mc.

3.3.5.1.1.3 Transmitted Control Pulse Width - The pulse width at the half amplitude point is at least 0.7 usec. and not greater than 0.9 usec in duration.

3.3.5.1.1.4 Transmitted Control Pulse Rise Time - The rise time (10% to 90%) does not exceed 0.05 to 0.1 usec. and the fall time (90% to 10%) from maximum amplitude will be from 0.05 to 0.2 usec.

3.3.5.1.1.5 Transmitted Control Pulse Position - The leading edge of the control pulse occurs 2.0 $\pm$ 0.15 usec. after the leading edge of the first interrogation pulse, as measured at the half-amplitude points.

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3.3.5.2 Receiver Side Lobe Suppression Characteristics - The equipment shall respond to side lobe suppression transmissions as follows:

3.3.5.2.1 Pulse Amplitude Difference - The equipment shall be suppressed by an interrogation when the amplitude of the control transmission pulse is equal to or in excess of the level of the first interrogation pulse. The equipment shall reply to an interrogation when the amplitude of the first interrogation pulse exceeds the amplitude of the control transmission pulse by 9 db or more. The equipment may or may not reply within the 9 db level difference.

3.3.5.2.2 Suppression of Replies - Upon identifying a side lobe interrogation, an inhibiting gate shall be generated to prevent replies thereto. These functions shall be performed as specified over at least a 50 db range of signal levels above the specified minimum triggering level of 3.3.3.2.

3.3.5.2.3 Inhibiting Gate Length - The inhibiting gate, generated to inhibit coder-receiver-transmitter response to side lobes, shall be 35 $\pm$ 10 usec. It shall be capable of being regenerated for the full duration within 2 usec after the end of any suppression period.

3.3.5.2.4 Side Lobe Suppression Rate Limiter - The inhibiting gate circuit shall be interconnected with the AOC circuitry to provide the following:

(1) At a side lobe suppression rate up to 3000 per second there shall be no effect on the trigger level.

(2) At a side lobe interrogation rate in excess of 5000 per second, the trigger level shall be reduced by AOC action to limit the side lobe suppression rate to a maximum of 10,000 per second.

3.3.6 Transmitter Characteristics - The equipment shall respond to interrogations defined in 3.3.2.1 by transmitting pulse-train replies for each interrogation received, depending upon the mode of interrogation used and upon equipment control settings selected.

3.3.6.1 Transmitter Frequency - The transmitter shall be adjusted by the contractor for center frequency operation at 1090 $\pm$ 0.5 mc.

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3.3.6.1.1 Transmitter Frequency Drift - The transmitter frequency shall remain within ± 3.0 mc of 1090 mc for all combinations of service conditions, including

(1) Variations of interrogation rates to cause a duty cycle variation from 0.001 to 0.01

(2) An antenna mismatch which is such as to cause a standing wave ratio of 3.5 db (VSWR 1.5:1) with the indicent and reflected power at a phase relationship which causes a maximum reduction in power.

(3) Change from standby to full operation.

3.3.6.2 Transmitter Power Output - Under any combination of service conditions and reply codes with the equipment set at any mode of interrogation, and at a reply rate of 1200 per second unless limited by AOC action (3.3.3.8), the transmitter shall deliver 27 ± 3 db above 1 watt of RF power into a 51 ohm resistive termination at the antenna terminal of the equipment. The power output shall not be reduced to less than 24 db above one watt when the antenna mismatch is such as to cause a standing wave ratio of 3.5 db (VSWR of 1.5:1) with the incident and reflected waves at a phase relationship which causes a maximum reduction in power. Independent of the above-listed variations in power output, the power output shall not be reduced by more than 1 db when the reply rate is increased from 1200 to 2000 pulse trains per second and by not more than 2 db when further increased to 3500 pulse trains per second.

3.3.6.2.1 Power Output Variation Within a Group - The peak power of any pulse in a group of pulses specified in 3.3.7 shall not differ from any other pulse by more than 1 db.

3.3.6.3 R-F Tube Change - The equipment design shall permit the replacement of the transmitter tube without the necessity of altering equipment adjustments other than frequency tuning controls.

3.3.6.4 High Voltage Delay - Provisions shall be incorporated in the equipment to prevent the Transmitter from transmitting for a period of 75 ± 15 seconds after power has been switched to the unit. Filament, bias and high voltages may be applied as soon as power has been switched on.

3.3.6.5 Transponder Delay - The delay between the second pulse of any interrogation mode except Mode 4 received at the antenna terminal and the first reply pulse delivered at the antenna terminal shall be 3.0 ± 0.5 usec. This shall apply for interrogation signals from MTL to 50 db above MTL with pulses at nominal spacing. Delay variation between modes on which the transponder is capable of replying shall not exceed 0.2 usec.

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3.3.6.6 Range Jitter - Under conditions described in 3.3.6.5 with successive interrogations of equal amplitude the maximum jitter between successive replies shall not exceed 0.1 microsecond provided that the total number of interrogations does not exceed 2000. This applies for signal levels between 3 and 50 db above MTL.

3.3.7 Reply Characteristics - The transmitted replies as measured at the antenna terminal shall have the following characteristics:

3.3.7.1 Reply Pulse Characteristics

Duration:	0.45 ± 0.1 usec
Rise Time:	0.05 to 0.1 usec
Decay Time:	0.05 to 0.2 usec
Amp. Jitter:	5% or less

Note: The rise time may be less than 0.05 usec providing the actual spectrum resulting is shown to fall within an envelope of the worst case theoretical spectrum of a pulse having a 0.05 usec rise time. The envelope shall be a line smoothly connecting the harmonic peaks of the theoretical spectrum.

3.3.7.2 Mode 1 - For each pulse appearing at the Mode 1 decoder output, a pulse train shall be transmitted containing from none to a maximum of 5 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 usec. The information pulse spacing shall be in multiples of 2.9 usec from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 usec. The position where a sixth pulse would appear (17.4 ± 0.05 usec from the initial framing pulse) will not be used. From the specified 5 information pulses, a total of 32 different codes remotely selectable are available.

3.3.7.3 Mode 2 - For each pulse appearing at the Mode 2 decoder output, a pulse train shall be transmitter containing from none to a maximum of 12 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 usec. The information pulse spacing shall be in multiples of 1.45 usec from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 usec. The position where a seventh information pulse would appear (10.15 ± 0.05 usec from the initial framing pulse) will not be used. Facility for selecting any of the 4096 codes available shall be provided on the equipment front panel.

3.3.7.4 Mode 3/A - For each pulse appearing at the Mode 3/A decoder output, a pulse train shall be transmitted containing from none to a maximum of 12 information pulses plus 2 framing pulses which are spaced 20.3 ± 0.05 usec. The information pulse spacing shall be multiples of 1.45 usec from the initial framing pulse with a maximum allowable deviation from this spacing of ± 0.05 usec. The position where a seventh information pulse would appear (10.15 ± 0.05 usec from the initial framing pulse) will not be used. From the specified 12 information pulses, a total of 4096 different codes remotely selectable are available.

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3.3.7.5 Mode 4 - Mode 4 encoding is performed in an external computer.

3.3.7.6 Mode C - For each pulse appearing at the Mode C decoder output when the receiver-transmitter is connected to an altitude computer-transducer unit, a pulse train shall be generated containing 1 or more information pulses, plus 2 framing pulses spaced 20.3 usec and the SPI (Special Position Identification) pulse (paragraph 3.3.7.9). The information pulses shall be referenced from the first framing pulse in multiples of 1.45 usec with a maximum allowable deviation from this spacing of ± 0.05 usec. The positions where a seventh pulse (X) would appear (10.15 ± 0.05 usec from the initial framing pulse) and where a ninth pulse would appear (13.05 ± 0.05 usec from the initial framing pulse) will not be used. Whenever the thirteenth pulse position (18.85 ± 0.05 usec) is chosen, the SPI pulse shall also be generated at the output. The SPI pulse shall be spaced 24.65 ± 0.05 usec from the first framing pulse. Control leads shall be provided to accept digitized altitude information from the computer-transducer.

3.3.7.7 Emergency Replies, Modes 1, 2, and 3/A - When the receiver-transmitter is switched to Emergency, the coder shall generate multiple pulse trains as follows:

(1) Mode 1 - For each trigger pulse appearing at the Mode 1 decoder, the code in use will appear in the first train, followed by 3 sets of framing pulses with these pulses appearing at 24.65 ± 0.1, 44.95 ± 0.15, 49.30 ± 0.20, 69.60 ± 0.25, 73.95 ± 0.30 and 94.25 ± 0.35 usec.

(2) Mode 2 - For each trigger pulse appearing at the Mode 2 decoder output, the code in use will appear in the first train, followed by 3 sets of framing pulses, with the pulses spaced as for Mode 1.

(3) Mode 3/A - For each trigger pulse appearing at the Mode 3/A decoder output, the code number 7700 will appear in the first train, followed by 3 sets of framing pulses, with the pulses spaced as for Mode 1. Replies in Modes 1, 2 and 3/A shall be enabled regardless of the setting of the remote mode-select switches.

3.3.7.8 Emergency Replies, Modes 4 and C - The emergency function does not affect Modes 4 and C.

3.3.7.9 Identification of Position (I/P) - Provision shall be made for I/P operation. A time delay shall be provided so that when the I/P control lead is initially grounded (regardless of length of grounding period) by remote control, I/P action will be initiated and will continue for a pre-set period between 15 and 30 seconds. The I/P timing operation shall be capable of being

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reinitiated at any time. When the receiver-transmitter is switched to I/P, the coder shall generate coded replies with the following characteristics:

(1) Mode 1 - For each trigger appearing at the Mode 1 decoder output, the code in use shall appear twice. The second pulse train shall be spaced 24.65 ± 0.05 usec as measured between the leading edges of the first framing pulses of the two trains.

(2) Mode 2 - For each trigger pulse appearing at the Mode 2 decoder output, the code in use shall appear followed by a single pulse (SPI pulse) spaced 24.65 ± 0.05 usec from the first framing pulse.

(3) Mode 3/A - For each trigger pulse appearing at the Mode 3/A decoder output, the code in use shall appear followed by a single pulse (SPI pulse) spaced 24.65 ± 0.05 usec from the first framing pulse.

(4) Other Modes - The I/P function shall not affect the normal operation of Mode C and Mode 4.

3.3.7.10 Altitude Reporting Provisions - When a Mode C interrogation is decoded, the coder shall generate a pulse train as described in paragraph 3.3.7.6. The information pulses are remotely selected by an altitude computer-digitizer.

3.3.7.10.1 Voltage and Resistance Turn-off and Turn-on Limits for Altitude-Code Select Wires - Input connections shall be provided on the receiver-transmitter to accept Mode C reply code information from devices using either electronic keying or disc-brush encoder. Mode C control characteristics shall be as follows:

(1) A turn-off (information bit not present) shall occur when tested separately under both of the following conditions:

a. When a low-impedance voltage applied to the control lead is at least + 18.5 but not exceeding + 30 volts.

b. When an external resistance in excess of 100,000 ohms to ground is applied to the control lead.

(2) A turn-on (information bit present) shall occur when tested separately under both of the following conditions:

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a. When a low impedance voltage applied to the control lead is + 2.0 volts or less.

b. When an external resistance of 1500 ohms or less to ground is applied to the control lead.

(3) Steady-state current in any control lead shall not exceed 5.0 milliamperes when connected to ground.

3.3.7.10.2 Noise filtering on Mode C Leads - Suitable filtering shall be provided within the receiver-transmitter to limit the capacitance discharge current from an external encoder disc brush to 1.5 milliamperes or less when added to the steady state brush current. The use of inductive-type filtering shall not be permitted.

3.3.7.10.3 Source and Transient Voltages - The source of the energizing voltages for the coding circuits within the receiver-transmitter shall be such that the open circuit voltage on the control leads will not exceed 30 volts. Transient protection on this source voltage shall be provided so that irrespective of the transients which might appear on the input power leads to the receiver-transmitter, the peak inverse voltage on the control lead will not exceed 30 volts.

3.3.7.11 Auxiliary Trigger Operation - An input shall be made available with an input impedance of 90 ± 10% ohms for accepting external triggers with the following characteristics:

Duration	0.35 to 1.5 usec.
Amplitude	15 to 30 volts
Rise Time	0.1 usec or less
Decay Time	0.2 usec or less
Polarity	Positive
Pulse spacing	2.0 usec minimum

For each trigger received the equipment shall transmit a single reply pulse with characteristics described in paragraph 3.3.7.1.

3.3.8 Code Number Assignments - Every configuration of Mode 1, 2, and 3/A information pulses shall be assigned a code number. The following numbers shall be used:

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3.3.8.1 Mode 1 - Code numbers 00(00) through 73(00) shall designate the Mode 1 code. The applicable 12 information pulses shall be divided into 2 groups (A and B) with A containing the first 3 pulses and B containing the last 2 pulses. The 3 information pulses in group A shall be numbered in order serially (1, 2, and 4) and the 2 pulses in group B shall be numbered in order (1 and 2). The appropriate code number shall be obtained by adding the pulse numbers (1, 2, and 4) of the pulses present within each group and combining the 2 sums in order (A, B) to form the serial code numbers. Thus, code 03(00) consists of no pulses in group A, and pulses 1 and 2 of group B; code 51(00) consists of pulses 1 and 4 of group A, and pulse 1 of group B, etc.

3.3.8.2 Mode 2 - Code numbers 0000 through 7777 shall designate the Mode 2 code. The applicable 12 information pulses (it is not proposed to use the seventh (X) pulse position) shall be divided into 4 groups of 3 pulses. The individual pulses of each group shall be spaced 2.9 microseconds apart and numbered in order (1, 2, and 4). The position of the first pulse of each group, measured from the initial framing pulse, shall be as follows:

Group	Time From First Framing Pulse (Microseconds)
A	02.9
B	11.6
C	01.45
D	13.05

The appropriate code number shall be determined by adding the sum (1, 2, and 4) of the pulses present within each group. These sums shall then be combined in order (A, B, C, D) to form a 4-digit serial number for the particular code. For example, code 2435 will have information pulses at the 1, 3, 4, 9, 12, and 13 information pulse positions.

3.3.8.3 Mode 3/A - Code numbers 0000 through 7777 shall designate the Mode 3/A code. The same system shall apply to Mode 3/A as that described in 3.3.8.2.

3.3.8.4 Mode C - The hybrid reflected binary (Gray) code shall designate the Mode C code. The X and D_4 pulse positions shall not be used. The I/P and D_4 pulses shall be considered a single bit of information and both pulses shall be generated wherever D_4 is enabled.

3.3.9 Self Test - The unit shall provide circuitry to allow functional testing of the receiver sensitivity, receiver frequency, decoder, encoder, modulator, transmitter, power output, and transmitter frequency. The self test circuit shall be operated by a push-to-test switch on the front panel to indicate a go-no-go condition.

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3.3.9.1 Self Test Interrogator - A two pulse interrogation shall be generated for modes 2 and 3/A and shall be injected into the receiver through a directional coupler in the antenna lead. The interrogation shall be at an r-f frequency of 1030 + 0.250 mc. The signal level shall be adjustable from -66 to -74 dbm and shall be stable to an accuracy of ± 3 db.

3.3.9.2 Self Test Monitor - The r-f energy from the transmitter shall be applied to a 1090 mc cavity resonator filter and then detected in an amplitude sensitive circuit. The monitor shall provide a visual indication if the transmitter power output exceed a predetermined level to an accuracy of ± 4 db and if the transmitted r-f frequency is 1090 ± 4 mc.

3.3.10 External Suppression Output - The equipment shall generate a suppression pulse during the time of transmission which may be used to suppress other equipments. The suppression pulse shall be available at an output jack (which may be part of a main chassis connector) and have the following characteristics:

(1) Amplitude - 20 to 50 volts

(2) Polarity - positive

(3) Synchronization - The pulse shall reach 20 volts minimum by the time an R.F. output pulse has reached 10% of its detected amplitude.

(4) Duration - The suppression pulse shall remain at 20 volts or greater until the trailing edge of the last transmitted pulse has decayed to the 10% level. Suppression output shall also be less than 5 volts, 10.0 microseconds after the last framing pulse of the reply pulse train has fallen to 10 per cent level.

(5) Rise Time - Equal to or greater than 20 volts per microsec

(6) Decay Time - Equal to or greater than 10 volts per microsec

(7) System Impedance - The load impedance presented to the suppressor output shall be not more than 2200 ohms nor less than 300 ohms shunted by not more than 1850 pico-farads capacitance. Damage to the equipment shall not result from an "open" or "short" of the output jack.

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(8) Spurious Output - Neither the DC steady state output potential nor the instantaneous peak AC potential of spurious signals shall exceed 1.0 volt, either polarity, as measured with system loads as described above.

3.3.11 Mode 4 Operation - During Mode 4 operation the decoder-receiver-transmitter shall perform the following functions.

3.3.11.1 Mode 4 Trigger - Upon decoding the first four pulses of a Mode 4 interrogation (3.3.2.1) the equipment shall generate a trigger pulse. When terminated in 91 ohms ± 10% in parallel with 1200 micromicrofarads ± 10%, the trigger pulse shall have the following characteristics:

Amplitude:	2 to 20 volts
Polarity:	Positive
Rise Time:	0.1 microsecond, maximum
Decay Time:	1.0 microseconds, maximum
Duration:	0.5 to 3.0 microseconds
Undesired Signal Plus Quiescent d-c Voltage	From + 0.7 volt maximum to -0.7 volt minimum
Spacing:	The spacing of the Mode 4 trigger from the position of the fourth sync pulse in the Mode 4 video (3.3.11.2) shall be from plus 0.2 to plus 0.6 microsecond, nominally. This spacing shall not vary more than plus or minus 0.07 microsecond throughout the extreme operating conditions and at interrogation rates up to the turn down point of the Mode 4 rate limiter (3.3.11.4).

3.3.11.2 Mode 4 Video - The equipment shall route interrogation pulses to an external coder-decoder.

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When terminated in 91 ohms ± 10% in parallel with 1200 micromicrofarads ± 10%, the output shall have the following characteristics:

Amplitude:	1.5 to 5 volts
Polarity:	Positive
Duration:	0.4 to 0.6 microsecond
Rise Time:	0.1 usec maximum
Decay Time:	0.2 usec maximum
Undesired Signal Plus Quiescent d-c Voltage:	From + 0.7 volt maximum to -0.7 volt minimum
Pulse Train Droop:	The amplitude of any pulse in a 36 pulse group consisting of pulses spaced as close as 2 usec apart shall not differ from the amplitude of the first pulse of that group by more than 10%.
	The video will consist of four sync pulses followed by a vacant pulse followed by as many as 32 pulses. There will be an anti-interference pulse inserted between any consecutive vacant pulse spaces within the interrogation pulse train. The spacing from the leading edge of the first pulse of the video train to the leading edge of succeeding pulses of that train will be within plus or minus 0.1 microsecond of a multiple of two microseconds. The spacing of the anti-interference pulses will be plus or minus 0.15 microsecond of odd multiples of one microsecond from the leading edge of the fourth sync pulse.

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3.3.11.3 Mode 4 Reply Input - An input shall be provided for application of Mode 4 reply triggers to this equipment. The Mode 4 reply input pulses shall have the following characteristics when measured across 91 ohms ± 10%, in parallel with 100 micromicrofarads ± 10%:

Amplitude:	3 to 6.5 V
Polarity:	Positive
Duration:	0.3 to 0.7 us
Rise Time:	0.1 usec maximum
Decay Time:	0.25 usec maximum
Minimum Spacing:	1.6 usec
Quiescent d-c Voltage:	Between -4 and + 2 volts
Undesired signal from the external source:	± 0.7 volt maximum
Pulse Position of the first reply pulse:	Between 200 and 260 usec after the fourth pulse of a Mode 4 interrogation.

For each input trigger an output r-f pulse shall be generated having the characteristics described in 3.3.7.1.

In normal operation, Mode 4 replies will consist of a three-pulse group having a normal spacing of 1.8 microseconds between adjacant pulses. The total spacing between the first and third pulses will not differ from its nominal value of 3.6 usec by more than 0.2 usec.

3.3.11.4 Mode 4 Duty Limiter - This control shall limit the rate of decoding Mode 4 interrogation sync patterns by gradual reduction of triggering level. The threshold for reduction shall be adjustable between 1500 and 2500 Mode 4 enabling triggers per second. The maximum rate shall not exceed 120% of the rate at which reduction first begins if the interrogation signal is increased 50 db above minimum triggering level. If the interrogation contains an ISLS/5th bit or if the Mode 4 trigger is followed within 10 usec by a computer reset trigger (3.3.11.5.2), triggering level reduction shall not be counted for Mode 4 Duty Limiter purposes.

3.3.11.5 Mode 4 Sync Anti-Jam Operation

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3.3.11.5.1 Sync Anti-Jam Detection - If the interfering signal consists of randomly spaced pulses only and the interrogating signal is a Mode 1, 2, 3/A or C interrogation that is 10 db above the interfering signal, the equipment shall reply to at least 90 percent of the interrogations where the randomly spaced pulses are not recognized as normal or side lobe interrogations.

3.3.11.5.2 Mode 4 Computer Reset Trigger - A trigger shall be generated under the conditions described in paragraph 3.3.3.7.4(3). This trigger shall have the following characteristics when measured across 1000 ± 10% ohms:

Amplitude:	**6 to 10 volts**
Polarity:	**Positive**
Duration:	**0.3 to 10.0 usec**
Rise Time	**0.25 usec maximum**
Decay Time	**5.0 usec maximum**
Source quiescent DC voltage:	**0 to + 6 volts**

3.3.12 **Mode 4 Monitor - The unit shall provide circuitry to allow monitoring Mode 4 operation. When operating with an appropriate external coder-decoder and control unit, the equipment shall present aural and visual indications of the Mode 4 interrogation and reply activity.**

3.3.12.1 Mode 4 Audio Output - An audio output signal shall be generated if a Mode 4 trigger (3.3.11.1) is generated and a disparity (3.3.12.4) is not received within 200 microseconds and an external control lead designated "Audio Enable" is grounded.

3.3.12.1.1 Mode 4 Audio Signal Characteristics - The audio signal output shall have the following characteristics when measured across 600 ohms ± 10%:

Amplitude:	Internally adjustable from 0 to 5 volts peak
Duration:	100 to 400 microseconds
Frequency:	Pulse repetition frequency of Mode 4 triggers (3.3.11.1)

The audio output and audio return pin connections shall be electrically isolated from ground.

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3.3.12.2 Mode 4 Reply Indication - A reply indication signal shall be generated if four or more transmitted reply groups are generated from reply triggers (3.3.11.3) within a 1/30 second interval or the reply groups from reply triggers are generated at a rate greater than 50 per second for 0.25 second. The reply signal shall not be generated if transmitter reply groups from reply triggers are generated at a constant rate of less than ten per second. The reply indication signal characteristics are described in 3.3.13.

3.3.12.3 Caution Light Enable - Provisions shall be made to enable an external caution light whenever either of the following conditions exist:

(1) Zeriozed Mode 4 reply code is indicated by grounding of a control lead by the external Mode 4 computer.

(2) The receiver-transmitter has failed to reply to valid Mode 4 interrogations four or more times within a 0.033 second interval or to valid Mode 4 interrogations at a rate greater than 50 per second for 0.25 seconds. A failure shall be considered to have occurred when a Mode 4 sync pattern is recognized but not followed within 270 usec by either a Mode 4 reply indication (3.3.12.2) or a Condition No. 1 or No. 2 Disparity pulse (3.3.12.4). A disparity pulse Condition No. 3 (3.3.12.4) occurring at approximately 198 usec after the Mode 4 enabling trigger shall not inhibit the caution light. An exception to the "failure to reply" indication shall apply when the failure to reply to a valid Mode 4 interrogation occurs at a constant rate of less than 10 times per second. The lamp shall remain on for a period of from 2 to 5 seconds and shall automatically reset.

3.3.12.4 Disparity Input - A video input shall be provided for application of Mode 4 Disparity trigger pulses to the equipment. Characteristics of the Disparity pulse, when measured across 90 ± 10% ohms shall be as follows:

Amplitude:	3 to 6.5 volts
Polarity:	Positive
Duration:	0.3 to 1.0 usec
Rise Time:	0.15 usec, maximum
Fall Time:	0.5 usec, maximum
Load Quiescent DC Voltage:	Plus 2 to minus 4 volts

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Undesired signals:	Less than ± 0.7 volts
Negative overshoot caused by load:	Less than 1.5 volts
Timing from fourth sync pulse:	Condition No. 1 (ISLS) less than 4 usec Condition No. 2 (Sync A/J) 6 ± 1.0 to 74 ± 2 usec Condition No. 3 (Disparity) 198 ± 2 usec

3.3.13 Reply Indication Signal Characteristics - A reply indication signal is used to indicate Mode 4 reply activity. The equipment shall provide a signal having the following characteristics when terminated in 150 ohms ± 10%.

Amplitude:	23 to 33 VRMS
Duration:	Mode 4 Reply - 2 to 5 seconds

3.4 Detailed Requirements

3.4.1 General - The Coder-Receiver-Transmitter shall fulfill the following requirements:

3.4.1.1 Function - The receiver portion of this equipment shall receive R-F energy from the antenna, amplify and decode it and initiate the proper transmitted pulses. The transmitter portion shall generate R-F power and transmit it to the load specified herein.

3.4.1.2 Form Factor - The coder-receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shock-mounted equipment shelf which in turn is mounted in an aircraft. It shall be possible to remove the equipment from the shelf and to remove the dust cover using ordinary hand tools without requiring removal of any cable. The over-all dimensions shall not exceed 7-9/16 by 6-9/16 by 22-7/8 inches. The front plate which contains mounting and extracting provisions extends an additional 1-1/32 inches below the base of the unit.

3.4.1.3 Contents - The Coder-Receiver-Transmitter shall consist of the following assemblies and associated devices with the exception of the transmitter tube all active devices are solid state.

(1) Preselector-Mixer-Assembly - This unit shall contain the preselector, mixer, local oscillator, duplexer (to permit a common antenna for receiver and transmitter functions) and directional coupler.

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(2) IF Amplifier Assembly - This unit shall provide the necessary I-F amplifier, the second detector, AOC circuitry and other circuits necessary to perform the required functions.

(3) Decoder Assembly - The unit shall contain video amplifier, pulse width discriminator mode decoders and SLS detection and suppression circuitry.

(4) Encoder Assembly - This unit consists of three plug in panels. Two panels shall contain circuitry for modulating the transmitter, the four mode gate generators, timing and pulse coincidence circuits and the reply pulse generator. The other panel shall contain the circuitry required to generate the appropriate information pulses.

(5) Power Supply Assembly - This unit shall contain the low voltage power supply.

(6) Transmitter, High Voltage Supply Assembly - This unit shall contain the transmitter cavity and tube and a high voltage power supply

(7) Mode 4 Monitor Assembly - This unit shall contain the required circuitry to perform the Mode 4 monitor functions.

3.4.1.4 Controls

3.4.1.4.1 Primary Power Controls - The equipment shall incorporate control circuitry to accomplish the following when used with Control, Transponder Set C-6280(P) ( )/APX in accordance with MIL-C-38257:

(1) In the "Standby" position of the control, the filament voltages shall be rated value, however, no R.F. shall be transmitted.

(2) In the "Normal" position of the control, all voltages shall be applied, the modulator shall be enabled and the unit shall be capable of normal operation immediately.

3.4.1.4.2 Mode 2 Code Selection - Switches shall be accessible from the front panel of the coder-receiver-transmitter unit to select Mode 2 coding.

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3.4.1.4.3 Internal Controls - The following semi-fixed adjustable controls shall be accessible when dust covers are removed.

(1)	Transmitter Tuning	Transmitter-Assembly
(2)	Pre-Selector Tuning	Preselector-Mixer Assembly
(3)	Normal Triggering Level	IF Assembly
(4)	Low Triggering Level	IF Assembly
(5)	AOC Level Set	Decoder Assembly
(6)	SLS Level Set	Decoder Assembly
(7)	Mode 4 Audio Output Level Set	Mode 4 Monitor Assembly
(8)	Interrogation Level Set	Self Test Assembly
(9)	Mode 4 AOC Level Set	Decoder Assembly

In addition to internal controls, internal adjustments for proper equipment alignment must also be accessible.

3.4.1.5 External Finish - The equipment covered by this specification shall be painted with color No. 26173 of FED-STD-595. The bottom surface of the equipment shall not be painted. Mating surfaces of covers, chassis and other points requiring electrical contact are to be free of paint.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

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(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that arepresentative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractor's expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one early production equipment shall be selected and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports shwoing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature:	Ambient room
Humidity:	Ambient room
Vibration:	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. The equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes or in a rotary direction for 10 minutes. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094.

4.4.3.3 Accept-Reject Criteria - Figure 14 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured

Characteristic	Spec. Requirement	Acceptance Requirement
Receiver Center Frequency	1030 ± .5 mc	1030 ± 1.5 mc
Normal Receiver Triggering Level	-77 dbm	-74 dbm

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Decoder		
Accept Limits		
Mode 1	3 ± 0.2 usec	3 ± 0.2 usec
Mode 2	5 ± 0.2 usec	5 ± 0.2 usec
Mode 3/A	8 ± 0.2 usec	8 ± 0.2 usec
Mode C	21 ± 0.2 usec	21 ± 0.2 usec
Transmitter Frequency	1090 ± 3 mc	1090 ± 3 mc
Transmitter Power	27 ± 3 dbw	24 dbw
SIF Reply		
Pulse Characteristics		
Bracket Pulses	20.3 ± 0.05 usec	20.3 ± 0.05 usec
Mode 1 (5 pulses at 2.9 usec intervals)	± 0.05 usec	± 0.05 usec
Mode 2 (12 pulses at 1.45 usec intervals)	± 0.05 usec	± 0.05 usec
Mode 3/A (12 pulses at 1.45 usec)	± 0.05 usec	± 0.05 usec
Emergency	See para. 3.3.7.7	See para. 3.3.7.7
Mode 4 Trigger	See para. 3.3.11.1	See para. 3.3.11.1

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 500-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government

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inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests, reliability tests, or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting pre-production tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - This equipment is mainly intended to be used as part of the Integrated Electronic Central AN/ASQ-19, AN/ASQ-82 and AN/ASQ-88. In this application, it furnishes aircraft identification features to ground or air interrogations.

6.2 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Definitions

6.3.1 Full Firing - Transmission of replies or Mode 4 triggers for at least 90 percent of the interrogation rate.

6.3.2 Interrogation - An R-F pulse or a combination of pulses intended to trigger a Coder-Receiver-Transmitter.

6.3.3 Interrogation Mode - The envelope characteristics of the interrogation.

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6.3.4 I-R - Interrogator-Responsor.

6.3.5 Jitter - A momentary variation in pulse separation, expressed in microseconds.

6.3.6 Pulse Amplitude - The maximum voltage amplitude of the pulse or the R-F pulse envelope, excluding spike.

6.3.7 Pulse Amplitude Jitter - The variation of the instantaneous pulse amplitude of successive pulses caused by unintentional effects within the equipment, expressed in terms or precent of average instantaneous amplitude.

6.3.8 Pulse Decay Time - The interval of time, expressed in microseconds, required for the trailing edge of the pulse to decay from 90% to 10% of the pulse amplitude.

6.3.9 Pulse Duration - The interval of time, expressed in microseconds, between the points at which the amplitude of the pulse is 50% of the pulse amplitude.

6.3.10 Pulse Rise Time - The interval of time, expressed in microseconds, required for the leading edge of the pulse to rise from 10% to 90% of the pulse amplitude.

6.3.11 Pulse Separation - The interval of time, expressed in microseconds, between 50% pulse amplitude points on the leading edges of the pulse involved.

6.3.12 Pulse Valley Amplitude - The minimum RMS amplitude (or instantaneous amplitude in the case of a video pulse) which exists between 90% pulse amplitude points of the pulse.

6.3.13 Reply - An R-F pulse or pulse group transmitted by a coder-receiver-transmitter as a result of an interrogation.

6.3.14 Squitter - Undesired spontaneous firing of a coder-receiver-transmitter, usually caused by internal noise.

6.3.15 Coder-Receiver-Transmitter Delay - The interval of time, expressed in microseconds, between the last component R-F pulse of the interrogation and the first component R-F pulse of the reply, measured between the 50% of pulse amplitude points on the leading edges of the pulses involved.

6.3.16 Minimum Triggering Level - Triggering level is a term applied to the minimum amplitude of the input RF interrogation pulses which just cause full firing of the transmitter.

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6.3.17 SIF - All interrogation modes except Mode 4.

6.3.18 Coder - A circuit which establishes the SIF reply.

6.3.19 Decoder - A circuit which responds to a particular interrogation mode and rejects all others.

6.3.20 Echo Suppression - A form of transponder circuit operation requiring successive pulses to be nearly the same amplitude to cause triggering.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7. **The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.**

6.5 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.7 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.8 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of packaging and packing (See 5.1).

6.9 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2, unless specified in the contract or order.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-S-[illegible]40(WP) CODER-RECEIVER-TRANSMITTER KY-532(*)/ASQ

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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